2013 Homework 2

Slew, clock-tree delay, mobility and noise margins

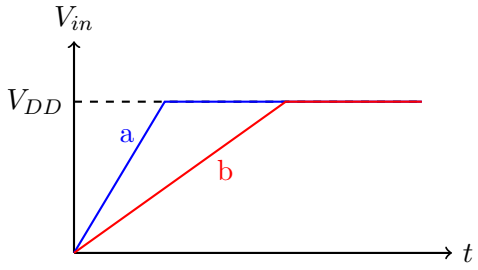
The original printed prompt is preserved at `../assignments/homework2.pdf`. Both handwritten answer photographs by student 100501015 were reviewed before derivation. They give the correct main conclusions; the qualifications and unit checks below make their assumptions explicit.

1. Faster rising input gives faster output fall

For the given inverter and fixed load,

$$C_L \frac{dV_{out}}{dt} = I_p - I_n.$$

The NMOS begins conducting when $V_{in} > V_{tn}$; the PMOS remains conducting while $V_{in} < V_{DD} - |V_{tp}|$. With input waveform a, the input reaches strong NMOS overdrive sooner and turns the competing PMOS off sooner. Therefore the net load-discharge current is larger during the transition, and the output falls faster. Waveform b spends more time with a weaker NMOS and simultaneous PMOS pull-up. This explains the stated fall-time ordering.



The reference uses the halfway point as an informal transition marker. Actual device turn-on/off voltages depend on thresholds; neither transistor universally switches off exactly at $V_{DD}/2$. The statement concerns the specified inverter experiment, not an unconditional bound for every circuit.

2. Critical path of the printed clock tree

The output-C buffer is labeled $2\tau_g$, while the A, B and D buffers have delay τ_g. Use the prompt's $r = 0.05\,\Omega/\mu$m and $c = 30 \times 10^{-18}$ F/μm. For an isolated length l in μm,

$$\tau_w = \tfrac{1}{2} r c l^2 = 0.75 \times 10^{-18} l^2 \text{ s}.$$

The printed neglect criterion is $\tau_w \le 4\tau_g = 0.8$ ns, not $\tau_g/4$. A 200 μm segment gives 0.030000 ps; a 250 μm segment gives 0.046875 ps. Even the following whole-path estimates are negligible:

Path	Wire length (μm)	$\frac{1}{2}rcl^2$ (ps)	Gate delay (ns)
A to B	650	0.316875	0.4
A to C	400	0.120000	0.6
A to D	650	0.316875	0.4

Critical path: A to C; delay $3\tau_g = 0.6$ ns.

The handwritten answer obtains the same result and properly uses $4\tau_g$. A branched tree's exact Elmore delay includes downstream branch loading and unspecified gate capacitances; the isolated-wire formula is the stated course approximation. The problem explicitly permits neglecting these very small wire terms, so they are not added as an exact physical tree-delay prediction.

3. Mobility ratio at the switching point

At $V_{in} = V_{out} = V_{DD}/2$, both conducting devices are saturated in the ideal long-channel model. With common C_{ox}, equal currents give

$$\mu_n \cdot 2\,(V_{DD}/2 - V_{tn})^2 = \mu_p \cdot 4\,(V_{DD}/2 - |V_{tp}|)^2.$$

Hence the general relation is

$$\boxed{\frac{\mu_n}{\mu_p} = 2\left(\frac{V_{DD}/2 - |V_{tp}|}{V_{DD}/2 - V_{tn}}\right)^2.}$$

Under the usual implicit matched-threshold assumption $V_{tn} = |V_{tp}|$,

$$\boxed{\mu_n/\mu_p = 2,}$$

as in the supplied answer. Unequal threshold magnitudes or oxide capacitances would prevent the two midpoint voltages alone from determining mobility uniquely.

4. Design for $NM_H > NM_L$

$$NM_H = V_{OH} - V_{IH}, \qquad NM_L = V_{IL} - V_{OL}.$$

With matched threshold magnitudes and nearly ideal rail outputs, make the NMOS pull-down stronger than the PMOS pull-up:

$$\boxed{\beta_n > \beta_p, \quad \mu_n(W_n/L_n) > \mu_p(W_p/L_p).}$$

Increasing W_n/L_n or decreasing W_p/L_p shifts the voltage-transfer curve left. The inverter accepts a lower input voltage as high, reducing V_{IH} and increasing NM_H; it also reduces V_{IL} and thus NM_L. If $\mu_n = 2\mu_p$ and lengths are equal, require $W_p < 2W_n$ for this strength ordering. A stronger PMOS would shift the characteristic in the opposite direction.

β_n/β_p	NM_L (V)	NM_H (V)
0.25	0.984360	0.603557
0.5	0.889328	0.688420
1	0.787500	0.787500
2	0.688420	0.889328
4	0.603557	0.984360

The table is an illustrative square-law check at $V_{DD} = 1.8$ V and equal threshold magnitudes 0.45 V. It solves the $dV_{out}/dV_{in} = -1$ boundaries; these are not values specified by the assignment or measured with CIC018.

Reference decisions. Retain the supplied faster-slew explanation, A-to-C critical path, mobility ratio 2 and $\beta_p < \beta_n$ sizing. Use threshold-dependent current balance instead of treating resistance matching as a universal transistor identity. Distinguish the transfer curve's switching point from the formal noise-margin slope boundaries.

Verification. `python verify.py` checks three clock paths and five ideal noise-margin examples. Results are in `clock_tree.csv`, `noise_margin.csv` and `verification.json`. These four questions are analytical; the prompt does not request a SPICE experiment.